CALCUL NUMERIC

30% act. laborator

3 sub t
12 prb } ex - 70%

Curs 1

1. Interpolare
2. Formule de cvadratură
3. Rezolvarea de ec.
4. Metode numerice
5. Rezolvarea aprox a ec diferențiale
6. Funcții radiale
7.

Bibliografie

1. Calcul numeric - Ioan
2. Introducere în metode numerice Atchinson
3. Calcul numeric - Coman Gh.

Interpolarea funcțiilor

$f : D \subset \mathbb{C} \to \mathbb{C}$, $z_0, z_1, \dots, z_n \in D$ (cunoaștem val) $f(z_k)$, $k = \overline{0,n}$

? aproximarea $f(z)$, $z \in D$, $z \neq z_k$

Lagrange

Fiind date $n+1$ pct distincte z_k, $k=\overline{0,n}$, $z_k \neq z_i$, $i \neq k$, $z_k \in D$, să se găsească un polinom P a.î. $P(z_k) = f(z_k)$ $\forall k = \overline{0,n}$

$Q(z) = P(z) + (z-z_0)\dots(z-z_n)H(z)$

Def Fiind date pct distincte $z_k \in D$, $k=\overline{0,n}$ s.n polinom de interpolare Lagrange polinomul de grad minim care verifică proprietățile $P(z_k) = f(z_k)$, $k=\overline{0,n}$ $\Rightarrow$ p interpolează fct f în pct z_k.

T $\exists$ un unic polinom Lagrange care interpolează fct f în pct z_k. El se notează $P(z) = L_n(f; z_0, \dots z_n)(z) = \sum_{k=0}^{n} l_k(z) f(z_k)$,

$$l_k(z) = \frac{l(z)}{(z-z_k)\, l'(z_k)}, \quad l(z) = \prod_{k=0}^{n} (z-z_k)$$

Dem a) Unicitatea

Pp $\exists P_1, P_2$ a.î. grad minim ... grad $P_i \le n$, $i=1,2$

$-P_1(z_k) = P_2(z_k) = f(z_k)$, $k=\overline{0,n}$

Fie $R(z) = P_1(z) - P_2(z)$, grad $R \le n$

$R(z_k) = P_1(z_k) - P_2(z_k) = f(z_k) - f(z_k) = 0$, $k=\overline{0,n}$

$\Rightarrow R = 0 \Leftrightarrow \underline{P_1 = P_2}$

$(z-z_0)\dots(z-z_n)' = $

$= (z-z_1)\dots(z-z_n) + (z-z_0)\dots(z-z_2)\dots$

$$l_k(z) = \frac{l(z)}{(z-z_k)(z_k-z_0)(z_k-z_1)\dots(z_k-z_{k-1})(z_k-z_{k+1})\dots(z_k-z_n)}$$

$$= \frac{(z-z_0)\dots(z-z_{k-1})(z-z_{k+1})\dots(z-z_n)}{(z_k-z_0)\dots(z_k-z_{k-1})(z_k-z_{k+1})\dots(z_k-z_n)}$$

$l_k(z_i) = \delta_{ki}$

$$P(z_i) = \sum_{k=0}^{n} l_k(z_i) f(z_k) = \sum_{k=0}^{n} \delta_{ki} f(z_k) = \delta_{ii} f(z_i) = f(z_i), \forall i=\overline{0,n}$$

Consecințe

$Q \in \Pi_n$ (grad $Q \leq n$) $z_i \in \mathbb{C}$, $i=\overline{0,n}$

$L_n(Q; z_0, z_1, \dots, z_n) = Q$ (din cauza unicității)

- $f(z) = \frac{Q(z)}{R(z)}$ grad $Q \leq$ grad $R \neq n+1$

$R(z_i) = 0$, $i=\overline{1,n}$

$$f(z) = \frac{L_n(Q, z_0, \dots, z_n)(z)}{A \cdot l(z)} = \frac{\sum_{i=0}^{n} \frac{l(z)}{l'(z_i)} \frac{1}{z-z_i} Q(z_i)}{R(z)} = \sum_{i=0}^{n} \frac{Q(z_i)}{R'(z_i)} \frac{1}{z-z_i}$$

coef.

$$= \sum_{i=0}^{n} \frac{1}{z-z_i} \operatorname{Rez}(f, z_i)$$

- caz particular:

$Q(z) = z^k$, $k \in \{0, 1, \dots, n\}$

$$\sum_{i=0}^{n} \frac{l(z)}{(z-z_i)\, l'(z_i)} z_i^k = z^k$$

$Q(z)$ $Q \in \Pi_{n+1}$, $a_{n+1} \neq 0$

$Q(z) = a_{n+1} z^{n+1} + a_n z^n + \dots + a_0$

$Q(z) - L_n(Q, z_0, \dots, z_n)(z) = a_{n+1}\, l(z)$

$Q(z) = L_n(Q, z_0, \dots, z_n)(z) + a_{n+1}\, l(z)$

Formula de recurență a lui AITKEN

$$\mathcal{L}_{n-1}^{\langle k\rangle}(z) = L_{n-1}(f; z_0, z_1, \dots, z_{k-1}, z_{k+1}, \dots, z_n)$$

Fie i, k, $i \neq k$, $i, k \in \{0, \dots, n\}$

$$\underbrace{L_n(f; z_0, \dots, z_n)(z)}_{\text{grad } n} - \mathcal{L}_{n-1}^{\langle k\rangle}(z) = A(z-z_0)\dots(z-z_{k-1})(z-z_{k+1})\dots(z-z_n) \,/\, (z-z_k)$$

$$L_n(f; z_0, \dots, z_n)(z) - \mathcal{L}_{n-1}^{\langle i\rangle}(z) = A(z-z_0)\dots(z-z_{i-1})(z-z_{i+1})\dots(z-z_n) \,/\, (z-z_i)$$

(-)

$$\Rightarrow (z_i - z_k) L_n(f; z_0, \dots, z_n) = (z-z_k) L_{n-1}^{\langle k\rangle} - (z-z_i) L_{n-1}^{\langle i\rangle}(z)$$

$$\boxed{L_n(f; z_0, \dots, z_n)(z) = \frac{(z-z_k) L_{n-1}^{\langle k\rangle}(z) - (z-z_i) L_{n-1}^{\langle i\rangle}(z)}{z_i - z_k}}$$ Formula lui Aitken

schema de obținere a polinoamelor:

$$\begin{array}{lll}
f(z_0) & & \\
 & > L_1(f; z_0, z_1) & \\
f(z_1) & & > L_2(f; z_0, z_1, z_2) \\
 & > L_1(f; z_1, z_2) & \\
\vdots & & \vdots \\
 & > L_1(f; z_{n-1}, z_n) & \vdots \\
f(z_n) & &
\end{array}$$

Def Fiind date pct distincte $z_0, z_1, \dots, z_n$ $f: D \to \mathbb{C}$, $z_i \in D$

S.n diferență divizată de ordin n a fct f pe pct $z_0, z_1, \dots, z_n$ coef lui z^n din polinomul de interpolare a lui Lagrange pe pct z_k

s.not $[z_0, z_1, \dots, z_n; f]$

$$\mathcal{L}_n(f; z_0,\dots,z_n)(z) = \sum_{k=0}^{n} \frac{(z-z_0)\dots(z-z_{k-1})(z-z_{k+1})\dots(z-z_n)}{(z_k-z_0)\dots(z_k-z_{k-1})(z_k-z_{k+1})\dots(z_k-z_n)} f(z_k)$$

$$\Rightarrow \boxed{[z_0,z_1,\dots,z_n; f] = \sum_{k=0}^{n} \frac{f(z_k)}{l'(z_k)}}$$

$$\mathcal{L}_m(f; z_0,\dots,z_m) = \frac{(z-z_k)\mathcal{L}_{m-1}^{<k>}(z) - (z-z_i)\mathcal{L}_{m-1}^{<i>}(z)}{z_i - z_k}$$

$$\boxed{[z_0,z_1,\dots,z_n; f] = \frac{[z_0,z_1,\dots,z_{k-1},z_{k+1},\dots,z_n; f] - [z_0,z_1,\dots,z_{i-1},z_{i+1},\dots,z_n; f]}{z_i - z_k}}$$

$$\boxed{[z_0,z_1,\dots,z_n; f] = 2\pi j \sum_{k=0}^{n} \operatorname{Rez}\left(\frac{f}{l(z)}; z_k\right) \frac{1}{2\pi j} = \frac{1}{2\pi j} \int_{C} \frac{f(z)}{l(z)}\, dz}$$

curbă simplă închisă care conține în int pct $z_0,\dots,z_n$

$[z_0,z_1,\dots,z_n; z^k] = ?$, $k=0,1,\dots,n,n+1,n+2$

$$[z_0,\dots,z_n; z^k] = \begin{cases} 0, & k \le n-1 \\ 1, & k=n \\ & \end{cases}$$

- $z^{n+1} - \mathcal{L}_n(z_0,\dots,z_n; z^{n+1})(z) = (z-z_0)(z-z_1)\dots(z-z_n)$

$-[z_0,z_1,\dots,z_n, z^{n+1}] = -(z_0+z_1+\dots+z_n)$

$$[z_0 z_1,\dots,z_n, z^{n+1}] = \sum_{k=0}^{n} z_k$$

- $z^{n+2} - \mathcal{L}_n(z^{n+2}, z_0,\dots,z_n)(z) = (z-\alpha)(z-z_0)\dots(z-z_n)$

0 coef lui z^{n+1} $= -\alpha - \sum_{k=0}^{n} z_k \Rightarrow \alpha = -\sum_{k=0}^{n} z_k$

coef lui z^n

$-[z_0,z_1,\dots,z_n; z^{n+2}] = \alpha \cdot \sum_{k=0}^{n} z_k + \sum_{i<k} z_i z_k$

$= -\left(\sum_{k=0}^{n} z_k\right)^2 + \sum_{i<j} z_i z_k$

$[z_0,z_1,\dots,z_n; z^{n+2}] = (\sum_{k=0}^{n} z_k)^2 - \sum_{i<k} z_i z_k$

$$[z_0, z_1, \dots, z_n; z^{n+2}] = \left(\sum_{k=0}^{n} z_k\right)^2 + \sum_{i<k} z_i z_k = \sum_{k=0}^{n} z_n^2 + \sum_{i<k} z_i z_k$$

$$\sum_{a_0+a_1+\dots+a_n=k} z_0^{a_0} \cdot z_1^{a_1} \cdot z_n^{a_n}$$

$f: I \subset \mathbb{R} \to \mathbb{R}$ $\quad x_i \in I,\ i = \overline{0,n},\ f \in C^{n+1}(I)$

$R_n(f)(x) = f(x) - L_n(f; x_0, x_1, \dots, x_n)(x)$

Fie $x = x_{n+1}$, $x_{n+1} \neq x_i$, $i = \overline{0,n}$

$L_{n+1}(f; x_0, x_1, \dots, x_n, x_{n+1})(x) - L_n(f; x_0, x_1, \dots, x_n)(x) = [x_0, x_1, \dots, x_{n+1}; f](x-x_0)\dots(x-x_n)$

$f(x_{n+1}) - L_n(f; x_0, \dots, x_n)(x_{n+1}) = (x_{n+1} - x_0)\dots(x_{n+1} - x_n)[x_0, x_1, \dots, x_{n+1}; f]$

$\boxed{R_n(f)(x) = l(x)\,[x_0, x_1, \dots, x_n, x; f]}$ } restul găsit cu ajutorul diferenței divizate

Formula de medie a lui Cauchy pt diferența divizată

[T] Dacă $f \in C^{n}(I)$ și dacă $x_i \in I$ distincte, $i = \overline{0,n}$ atunci $\exists$ un pct $\theta \in (\min(x_i), \max(x_i))$ a.î. $\boxed{[x_0, x_1, \dots, x_n; f] = \frac{f^{(n)}(\theta)}{n!}}$

$[x_0, x_1; f] = f'(\theta)$

$\boxed{\frac{f(x_1) - f(x_0)}{x_1 - x_0} = f'(\theta)} \to$ (Lagrange)

dem

$L_n(f; x_0, \dots, x_n)(x) = a_0 + a_1 x + \dots + a_n x^n$

$$\begin{cases} a_0 + a_1 x_0 + \dots + a_n x_0^n = f(x_0) \\ \dots \\ a_0 + a_1 x_n + \dots + a_n x_n^n = f(x_n) \\ a_0 + a_1 x + \dots + a_n x^n = L_n(f)(x) \end{cases}$$

$$\begin{vmatrix} 1 & x_0 & \dots & x_0^n & 0+f(x_0) \\ - & - & - & & \\ 1 & x_n & & x_n^n & 0+f(x_n) \\ 1 & x & & x^n & L_n(x)+0 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & x_0 & \dots & x_0^n & 0 \\ - & - & - & - & - \\ 1 & x_n & & x_n^n & 0 \\ 1 & x & & x^n & L_n(x) \end{vmatrix} + \begin{vmatrix} 1 & x_0 & x_0^n & f(x_0) \\ - & - & - & - \\ 1 & x_n & x_n^n & f(x_n) \\ 1 & x & x^n & 0 \end{vmatrix} = 0$$

$$L_n(f)(x)\, V(x_0, x_1, \dots, x_n) = - \begin{vmatrix} 1 & x_0 & x_0^n & f(x_0) \\ \dots & & & \\ 1 & x_n & x_n^n & f(x_n) \\ \cancel{1} & \cancel{x} & \cancel{x^n} & \cancel{0} \end{vmatrix}$$

$$[x_0, x_1, \dots, x_n; f]\, V(x_0, \dots, x_n) = - \begin{vmatrix} 1 & x_0 & \dots & x_0^{n-1} & f(x_0) \\ - & - & - & - & - \\ 1 & x_n & & x_n^{n-1} & f(x_n) \end{vmatrix}$$

$$[x_0, x_1, \dots, x_n; f] = - \frac{\begin{vmatrix} 1 & x_0 & \dots & x_0^{n-1} & f(x_0) \\ - & - & - & - & - \\ 1 & x_n & \dots & x_n^{n-1} & f(x_n) \end{vmatrix}}{V(x_0, \dots, x_n)}$$